

A Context-Free Fixation-Priority Heuristic: A Saliency Heatmap Based on Contour Density and Chromatic Discrepancy

— with an application to video keyframe extraction

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Abstract

This report presents a visual saliency computation method that requires no training, no semantic prior, and only linear time per frame. The method deliberately uses no task prior, no object knowledge, and no contextual information; it derives a saliency heatmap solely from the multiplicative combination of two statistical cues — the spatial clustering of local high-frequency energy (structural rarity) and the chromatic deviation from the global mean color (color rarity) — and outputs fixation-priority candidate regions. The central claim of this work is deliberately narrow: the algorithm does not model human attention in full, but rather the earliest, semantics-free stage of the visual pathway, i.e., a lower bound on the first fixation of an observer with no expectation and no task. Qualitative observation shows that the regions output by the algorithm are not always the first location humans fixate, yet they reliably fall within the priority fixation set; this report adopts that property as the operational definition of validity. The heatmap is further used as a semantics-free within-frame measure of where it is worth looking, and is combined with inter-frame differencing and a deduplication stage into a complete video keyframe extraction pipeline. Unusually for reports of this kind, an entire section is devoted to eight known deficiencies of the algorithm, each with its mechanism, impact, and suggested remedy — not as an appendix of confessions, but as a precondition for the correct use of the method.

Keywords: visual saliency; bottom-up attention; semantics-free heuristic; Laplacian energy; contour density; keyframe extraction

1 Introduction

1.1 Problem Statement

Given a single image, this report attempts to answer the following question: when a person sees the image for the first time in a completely unprepared state, with no expectation whatsoever, where is the gaze most likely to land first? Three key terms in this formulation require strict definition. First, "unprepared" excludes task-driven inspection: the observer is not searching for any particular target but passively receiving the image, which rules out all top-down attentional mechanisms. Second, "priority" refers to estimating the head of the fixation distribution — the candidates for the first few saccades — rather than

ranking all fixation points. Third, "most likely" indicates that the output is a statistical tendency rather than a deterministic prediction: the algorithm identifies statistically salient regions and guarantees no point-wise agreement with the actual fixation sequence of any particular observer.

The original formulation by the author of the algorithm was: to imitate where a person, in a state of complete unpreparedness, would most likely focus first when directly looking at an object or an image. After the refinement above, this formulation constitutes the complete problem definition of this report, and all subsequent design decisions and criticisms are developed against this baseline.

1.2 Design Stance: Deliberately Abandoning Semantics

Saliency research has historically followed two routes. The bottom-up route (Itti-Koch-Niebur [1]; Achanta et al. [2]; Hou and Zhang [3]) drives saliency computation purely from image statistics, without asking what is depicted in the scene; the top-down route, represented by contemporary deep models, relies on semantic priors and large-scale annotated data. This report chooses the former, and goes further than most classical work — even implicit semantic biases such as a face prior are not introduced. Both the benefits and the costs of this stance must be stated together.

There are four benefits. First, cross-domain generality: natural photographs, animation, screenshots, surveillance frames, and remote-sensing imagery all work with one fixed parameter set, without depending on weights trained on any particular domain and therefore without out-of-distribution failure. Second, zero training and zero datasets: there is no training-set bias to inherit and no annotation cost. Third, full explainability: any output region can be traced step by step along the pipeline to show exactly why it scored highly, a property that deep saliency models cannot provide. Fourth, the computational cost is linear in the number of pixels, implemented entirely with array operations, with no GPU dependency and deterministic, reproducible results.

There are likewise three costs, which must be declared in advance. First, the algorithm cannot understand what an object is: targets that are semantically important yet statistically flat (for example, a flat red flag lying on snow) may be systematically underestimated. Second, the algorithm uses no viewing history: human fixations after the first saccade are influenced by its outcome, and no temporal modeling is performed here. Third, the algorithm outputs a statistical baseline rather than a complete model of human fixation. The position of this report is that these costs are not defects but necessary consequences of the context-free design goal; if an application requires semantics, the correct approach is to stack a semantic layer on top of this algorithm's output rather than to mix semantic assumptions into the low-level computation.

1.3 The Boundary of the Claim to Emulate Human Vision

Strictly speaking, even the human bottom-up saliency map is modulated by semantics: face preference, viewing history, and task hints all reshape the fixation distribution. This report therefore does not claim to replicate the human eye. What it emulates is the earliest stretch of the visual pathway — subcortical through early primary visual cortex, before semantics participates — which computes statistical encodings of local contrast, spatial frequency, and color opposition. The precise claim is: this algorithm computes the statistical-prior component of the human fixation distribution, namely the lower bound on fixation tendency determined by the physical structure of the image itself before any semantic processing occurs. Any criticism of the form "it does not behave like a human eye" that points at the semantic level therefore does not refute this algorithm, because the algorithm never claimed to cover that level.

1.4 Structure of This Report

Section 2 surveys related work and locates the present algorithm among them; Section 3 describes the algorithm; Section 4 enumerates eight known deficiencies in strict terms, explaining for each why it holds and what it affects; Section 5 presents the qualitative observations and discusses the limitations of the current validity evidence; Section 6 develops the application to video keyframe extraction and argues why the semantics-free design fits that scenario; Section 7 concludes and lists future work.

2 Related Work and the Position of This Algorithm

Itti, Koch, and Niebur [1] established the classical framework for bottom-up saliency: feature channels (luminance, color opposition, orientation) extract center-surround contrast over a multi-scale pyramid, which is combined through competitive normalization into a saliency map, followed by a winner-take-all mechanism that selects attentional foci. The FT method of Achanta et al. [2] defines saliency directly as the distance from each pixel to the global mean color in Lab space, with band-pass filtering to suppress high-frequency noise, and remains the simplest effective representative of the color-rarity route. The spectral residual method of Hou and Zhang [3] removes statistical redundancy from the amplitude spectrum, likewise belonging to the semantics-free, training-free family.

The relationship of the present algorithm to these works must be stated honestly: its color channel is nearly isomorphic in structure to FT [2] (the differences are that it operates in RGB space and applies strong Gaussian blobbing; see Sections 4.3 and 4.4 for critiques of these choices), and its overall architecture is an extremely reduced single-scale variant of the Itti framework [1], with feature extraction, channel combination, and focus selection all drastically simplified. The incremental contributions of this report are two: first, the specific combination of edge density as the base channel with color rarity as a multiplicative gain,

whose AND semantics and its suppression of false positives on uniform color patches are analyzed in Section 3.4; second, the actual application of this saliency measure to video keyframe extraction as an engineering scenario, with an explicit argument for why the design fits. This report does not claim to have invented saliency detection.

3 Algorithm Description

3.1 Pipeline Overview

The full pipeline is as follows. Except for the two PIL image-filtering steps noted, all operations are numpy array computations, and the total complexity is linear in the number of pixels.

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RGB to float
-> Discrete Laplacian energy C    (local high-frequency / edge strength)
-> 99th-percentile strong-contour mask (keeps only the sharpest 1% of pixels)
-> 16x16 block contour density D    (spatial clustering of edges; the only scale parameter)
-> Chromatic discrepancy E, Gaussian-blobbed into A (deviation from the global mean color)
->  $S = D \times (1 + \alpha \cdot A)$ ,  $\alpha = 4$  (multiplicative fusion; AND semantics)
-> Gaussian smoothing (sigma=10) + UnsharpMask (blobbing and regional contrast enhancement)
-> Normalized heatmap macro_map
-> 80th-percentile threshold + connected components; Top 3 by intra-region saliency mass
-> Pseudo-color rendering (blue -> cyan -> green -> yellow -> red)

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3.2 Structural Channel: Laplacian Energy and Block Density

The structural channel answers the question of where edges cluster. In the first step, for every pixel the difference between the pixel and the mean of its four neighbors (up, down, left, right) is computed, and the squares are summed over the RGB channels, yielding the contrast map C where d denotes the difference between the center pixel and the four-neighbor mean. This operator is mathematically equivalent to a rescaling of the standard four-neighbor Laplacian L (specifically $d = -L/4$), so C measures the sum of squared Laplacian energy. Two reasons motivate this choice over gradient-magnitude operators of the Sobel family: the Laplacian kernel is approximately isotropic, responding to points, ridges, corners, and edges of arbitrary orientation without directional bias; and the squaring eliminates sign cancellation, measuring energy rather than polarity. It should be noted that d is not a strict discrete Laplacian (which subtracts the sum, not the mean, of the neighbors), but the two differ only by a constant factor, which has no effect on the subsequent percentile thresholding or normalization.

In the second step, the 99th percentile of C is taken as a threshold, and only the top 1% of pixels by contrast are kept as the strong-contour mask. The direct consequence is that, regardless of image content, exactly 1% of all pixels enter the density statistics. The subsequent density therefore measures the relative clustering of edges within the image rather than their absolute abundance. This is a defensible design — saliency is a relative

quantity, and the human eye likewise performs relative comparisons — but its cost is analyzed separately in Section 4.7.

In the third step, the mask is partitioned into 16x16 blocks, the fraction of strong-contour pixels within each block is computed, and the block values are expanded back to full resolution, producing a piecewise-constant local edge-density field D . This operation is equivalent to box-filter downsampling of the binary mask followed by nearest-neighbor upsampling. Note that 16 is the only scale parameter in the entire algorithm: a 300-pixel-wide text signboard and a stretch of forest may receive similar density values at this scale; the meaning and limitations of this fact are discussed in Section 4.5.

3.3 Color Channel: Chromatic Discrepancy

The color channel answers the question of where colors deviate from the environment. The Euclidean distance from each pixel to the global mean color is computed and normalized, then smoothed with a Gaussian of radius 15 to obtain the color-anomaly map A . The purpose of the blur is not denoising but a change in the semantics of the evidence: isolated color outliers are diluted by the blur, while coherent regions of anomalous color are preserved. This channel is nearly isomorphic to the core measure of FT [2], and no novelty is claimed for it here; critiques of the RGB metric and of the global-mean reference are given in Sections 4.3 and 4.4.

3.4 Channel Combination: Multiplicative AND Semantics

The two channels are combined as $S = D \times (1 + \alpha \cdot A)$ with $\alpha = 4$. Because the normalized D lies in the interval from 0 to 1, this combination carries explicit logical-AND semantics: wherever the density is zero, the score is zero regardless of how anomalous the color is; edge-dense regions with ordinary color receive moderate scores; only regions that are both structurally complex and chromatically outstanding receive high scores. This semantics is deliberate: it structurally suppresses false positives on uniform color patches (which contain no strong contours, so their density approaches zero). The price is the systematic suppression of large internally-flat chromatic targets, discussed in Section 4.6.

3.5 Blobbing, Enhancement, and Top-3 Region Selection

The combined map is first smoothed with a Gaussian of $\sigma = 10$ to connect scattered high responses into regions, and then sharpened with an UnsharpMask (equivalent to $S + 1.5 \times (S - G(S))$, radius 2, threshold 3) to crispen region boundaries. These are engineering steps that do not alter the theoretical structure. Finally, an 80th-percentile threshold binarizes the map, connected-component analysis is performed, and the top three regions are selected by their total intra-region saliency mass. This selection mechanism is a discretized replacement for the winner-take-all mechanism of the Itti framework: winner-take-all emulates the saccade-by-saccade shifting of the attentional focus, whereas Top-3 directly yields the head

candidate set, which for the present goal of a priority fixation set is both more fitting and simpler to implement. The pseudo-color mapping follows a piecewise linear blue-cyan-green-yellow-red scheme, monotonically aligned with saliency.

4 Known Deficiencies and Limitations

This section enumerates eight deficiencies of the current algorithm in strict terms. For each, the mechanism (why the problem holds), the consequence (what it affects), and a suggested remedy are given. Table 1 summarizes. The purpose of this section is to make the applicability boundaries explicit for anyone who cites or reproduces this algorithm.

4.1 Unpreconditioned Laplacian Is Hypersensitive to Compression Artifacts

The Laplacian operator amplifies high-frequency content approximately quadratically in the frequency domain, which makes it extremely sensitive to sensor noise and to the 8x8 block boundaries of JPEG compression. The high-frequency ringing that JPEG introduces at block boundaries can produce Laplacian responses exceeding those of genuine contours, so a substantial fraction of the strong-contour pixels selected by the 99th-percentile threshold may be compression artifacts rather than real structure, contaminating the density field downstream. The classical remedy is the LoG scheme of Marr and Hildreth [4]: apply a Gaussian smoothing with sigma of roughly 1 to 2 before taking the Laplacian. This is a one-line change, yet it aligns the algorithm with four decades of edge-detection theory and offers the highest benefit-to-cost ratio of all the suggested fixes.

4.2 Outlier Fragility of Min-Max Normalization

The algorithm uses $(x - \min)/(\max - \min)$ normalization in several places, which is the scheme most sensitive to outliers: if a tiny region of extreme density exists in the frame (a line of small text, for example), it monopolizes the maximum and everything else is compressed toward zero, where no amount of color gain can multiply it back. Two remedies exist: quantile normalization (capping at the 99th percentile) or compressing the dynamic range with $\log(1 + x)$ before normalizing. Neither changes the structure of the algorithm. The problem occurs in practice on scenes with small high-contrast targets, and a targeted test case is recommended.

4.3 RGB Euclidean Distance Is Not Perceptually Uniform

Human sensitivity differs across color channels: discriminability of luminance changes is far higher in the green channel than in the blue. An equal-weight RGB Euclidean distance therefore systematically underestimates green differences and overestimates blue ones. More importantly, this creates tension with this report's own claim: the report claims to emulate early-stage visual computation, yet the color coding of that stage (the opponent process with red-green and blue-yellow channels) is approximately perceptually uniform —

the equal-weight RGB distance does not even resemble the biological early pathway it invokes. Remedies: convert to CIELAB and take the Euclidean distance there (i.e., delta-E, the original choice of FT [2]), or implement a simplified opponent-channel representation. The cost is roughly three lines of code, and it is the single most necessary correction for the algorithm to be faithful to its own claim.

4.4 Systematic Bias of the Global-Mean Reference Color

Using the global mean color as the reference for chromatic discrepancy fails under certain compositions. A canonical counterexample: a red ball on a green lawn. The global mean is a muddy brown, and both the ball and the lawn — subject and background alike — lie far from it, so both receive high color-anomaly scores and the reference loses its discriminative power. It should be noted that this flaw also exists in FT [2] and is not unique to this work; nevertheless this algorithm can do better: estimate a background color from the pixels along the image border, or estimate rare colors directly from a color histogram instead of distances from the mean. The problem is most severe when the subject occupies close to half of the image or more.

4.5 Single-Scale Block Density

The 16x16 block size is the only scale parameter in the algorithm, meaning the density field is estimated at a single fixed spatial resolution. Biological vision is conspicuously multi-resolution: high acuity near the fixation point, dropping sharply in the periphery. The direct consequence of a single scale is that the internal structure of large objects and the overall contour of small objects cannot be characterized simultaneously: targets smaller than 16 pixels degenerate to single points in the density field, while targets much larger than the block contribute only their boundaries. Remedy: compute the density at three scales (8, 16, 32) and take a weighted sum. This is the cheapest step toward the multi-resolution character of early vision, and the most worthwhile ablation to run first.

4.6 Systematic Suppression of Large Flat Targets by the Multiplicative Fusion

While the AND semantics (Section 3.4) suppresses false positives on uniform color patches, it also suppresses a class of legitimate targets: large chromatic objects with flat interiors. A huge flat red flag on snow has near-zero interior Laplacian energy; only its boundary contributes strong-contour points, and after averaging into block density its score falls far below its visual salience. This is an inherent cost of the design trade-off rather than an implementation oversight: every AND-style combination has this property. If such targets matter for an application, an additive floor term for the color channel can be introduced ($S = D \times (1 + \alpha \cdot A) + \beta \cdot A$), but this simultaneously weakens the suppression of false positives on uniform patches — the two goals cannot both be maximized, and the trade-off must be made per application and reported honestly.

4.7 Relativity of the Percentile Mask

The 99th-percentile threshold forces the strong-contour pixels to constitute exactly 1% of every image, so the density measures the relative clustering of edges within a given image rather than absolute structural abundance. The consequence is that a minimalist image and a cluttered image have their density fields forcibly equalized onto the same scale, so scores are not comparable across images — only the ordering within a single image is meaningful. This is harmless for the goal of selecting regions within a single frame, but it constrains the application of comparing saliency across frames to pick keyframes (Section 6): cross-frame score comparison implicitly assumes comparable total edge mass per frame, and videos with abrupt editing-style changes must be handled with this caveat in mind.

4.8 No Temporal or Fixation-History Modeling

Human fixation is a temporal process: the outcome of the first saccade strongly influences where the second goes (inhibition of return, among other effects). This algorithm processes every frame independently and models no temporal dependency. For static attention-candidate estimation this is a reasonable simplification; for video applications it means that per-frame candidates carry no coherence constraint, and the candidate regions of adjacent frames may jump discontinuously. In the keyframe scenario this problem is partially masked by the downstream deduplication stage (Section 6.2), but any future extension to fixation-trajectory prediction would require introducing a temporal term.

No.	Issue	Primary impact	Suggested remedy	Priority
4.1	No low-pass preconditioning	Compression artifacts contaminate the strong-contour set and thus the density field	Gaussian smoothing (sigma 1-2) before the Laplacian (LoG)	High
4.2	Min-max normalization	Outliers monopolize the dynamic range; large areas collapse to zero	Quantile normalization or $\log(1+x)$ compression	High
4.3	RGB not perceptually uniform	Color metric contradicts the biological claim	Convert to CIELAB or opponent channels	High
4.4	Global-mean reference	Reference fails on two-subject scenes	Border-based background estimate or histogram method	Medium
4.5	Single-scale density	Multi-scale targets cannot be handled jointly	Weighted sum over 8/16/32 scales	Medium
4.6	AND suppresses flat targets	Large flat chromatic targets underscored	Additive floor term, per-application trade-off	Medium
4.7	Percentile relativity	Scores not comparable across images or frames	Fixed absolute threshold (requires experiments)	Low
4.8	No temporal modeling	Frame-wise candidates lack coherence	Mitigated by deduplication; add temporal term if extended	Low

Table 1 Summary of known deficiencies. Priority reflects the author's judgment of impact range versus fix cost.

5 Observations and a Discussion of Validity

5.1 Qualitative Observations

The current validation is qualitative in nature: the algorithm's outputs were observed manually on images of various types and compared against first impressions of human viewing. The test images cover different compositions including natural scenes and near-uniform textures (Figure 4); example outputs appear in Figures 1 and 3. The central observation is the following: the regions the algorithm marks as most salient are not always the first fixation of a human viewer, yet across all tested samples they reliably fall within the priority fixation set — the handful of locations a human would look at first (Figure 2 is an instance of "not the first choice, but within the set"). This report adopts that phenomenon as the operational definition of validity: the output does not match the human fixation sequence, but it provides a reliable superset estimate of the head of the fixation distribution.



Figure 1 Keyframe extraction output on a card-game video. Each cell is one frame; the timestamp and the trigger reason (focus / content / fixed / first) are annotated at the lower-left corner.

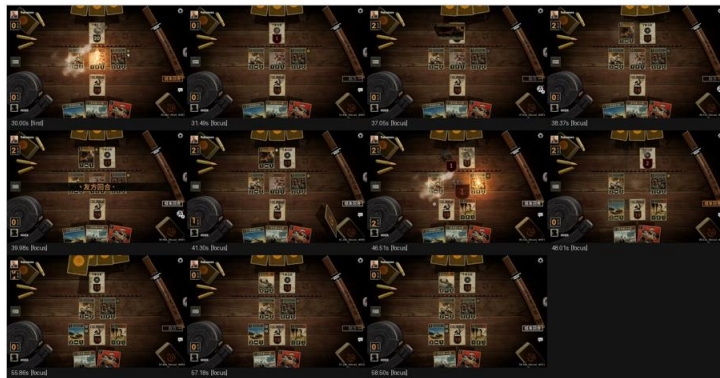


Figure 2 Extraction output on a later segment of the same video, showing the spatial distribution of candidate frames along a continuous time axis. Compared with Figure 1, the trigger density varies with frame content.



Figure 3 Extraction output on a live-action motion video. The input domain differs markedly from that of Figures 1 and 2, while all parameters are held constant, testing cross-domain generality.

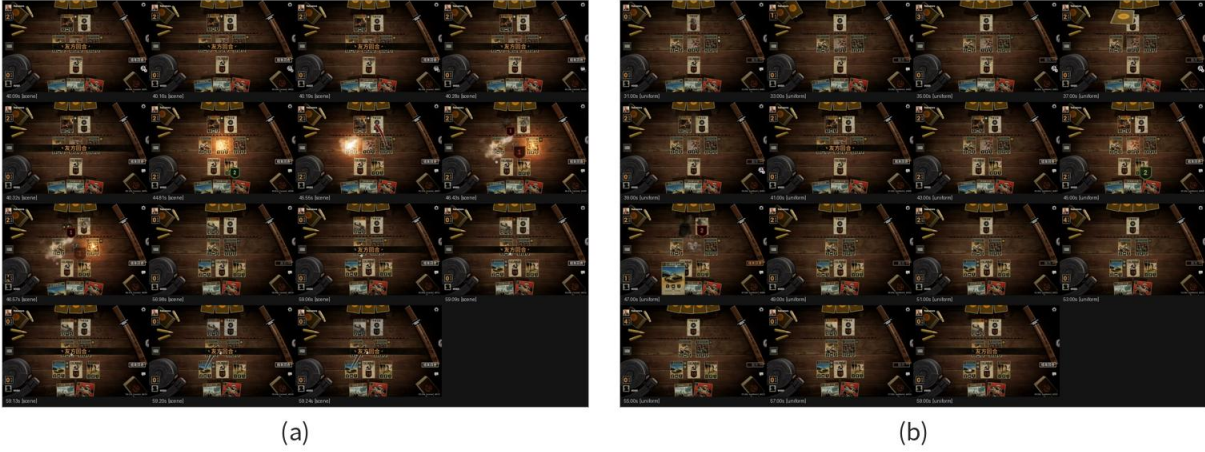


Figure 4 Two classes of test inputs: (a) frames from a card-game video; (b) frames from a game-UI video. Neither input carries the semantic priors typical of natural scenes, which tests the algorithm's behavior on inputs without a clear natural focus.

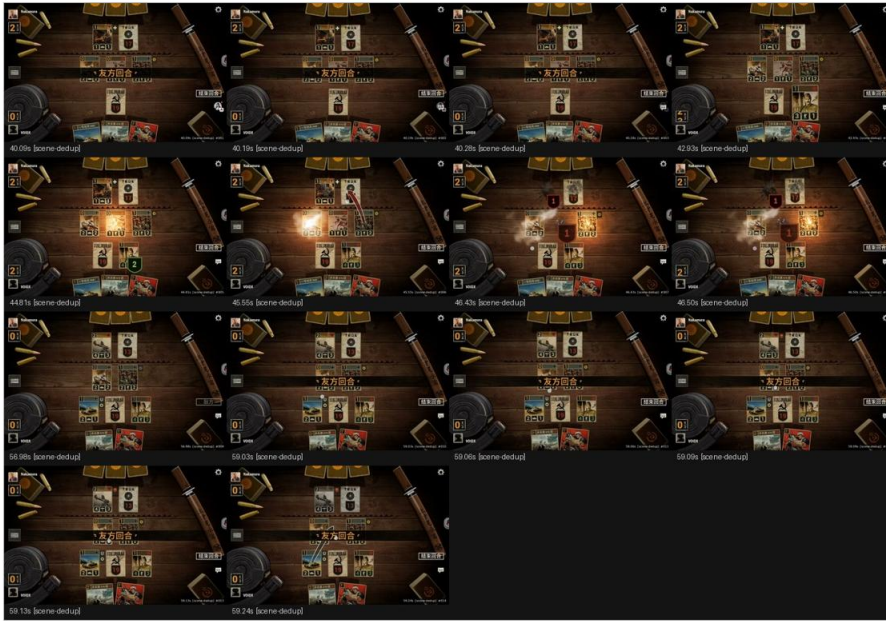


Figure 5 Deduplicated candidate-frame sequence. Frame-by-frame comparison shows that the visual differences between adjacent frames have been widened, indicating that the deduplication threshold is set reasonably and that no quota is wasted on near-identical frames.

5.2 Limitations of the Current Evidence and a Proper Validation Protocol

It must be stated strictly: the judgment that outputs "reliably fall within the priority fixation set" comes from the authors' subjective comparison, and its evidential grade is low, for three reasons. First, the judge is also the designer, so confirmation bias is present — hits are remembered more readily than misses. Second, the "human first impression" used for comparison is the introspective report of a single observer, whereas fixation distributions differ substantially across individuals. Third, no quantitative metric is employed: the boundary of the "priority set" is fixed post hoc, which risks circular reasoning.

A proper validation should follow this protocol: recruit a number of participants; record fixations under a free-viewing task with an eye tracker; compare the algorithm's output against the fixation-density map and report standard metrics such as AUC or NSS [5]; and

compare against at least one baseline (e.g., FT [2] or spectral residual [3]) and one deep model. Until eye-tracking is available, offline evaluation on a public eye-tracking dataset such as MIT1003 [6] is an adequate substitute. This report lists that validation as the highest-priority future work and declares explicitly that, until it is completed, the validity claim of this algorithm should be understood as "a design hypothesis with preliminary qualitative support," not as an established conclusion.

6 Application: Video Keyframe Extraction

6.1 Frame Scoring and Candidate Selection

The basic scheme for applying the saliency heatmap to keyframe extraction is: compute the heatmap for every frame of the video; use the energy of the heatmap (e.g., its mean or a high quantile) as a frame-level saliency score; and select candidate frames by score ranking. The relativity problem identified in Section 4.7 constrains this scheme: cross-frame score comparison implicitly assumes comparable total edge mass across frames, and videos with abrupt scene editing should be processed in segments or normalized within segments.

6.2 Deduplication Stage

Ranking frames by score tends to select clusters of visually near-identical consecutive frames. The deduplication stage uses a perceptual inter-frame distance (e.g., the mean absolute difference of downsampled grayscale frames) as its criterion and walks through the candidate frames in order: any frame whose distance to an already-selected frame falls below a threshold is discarded. The surviving set balances within-frame salience against between-frame difference, and serves as the engineering compensation for the absence of temporal modeling (Section 4.8).

6.3 Why the Semantics-Free Design Fits This Scenario

The requirements that keyframe extraction places on an algorithm differ from those of general saliency applications: what is needed is not a correct understanding of scene content, but a stable supply of usable candidates on videos of arbitrary domains, at negligible computational cost. This is precisely where the semantics-free design excels — the provenance of a video cannot be predicted in advance (it may span live footage, animation, and screen recordings), any model relying on domain-specific trained weights faces out-of-distribution failure, whereas this algorithm is plug-and-play across domains; and the total per-frame cost under a pure-array implementation is far below that of decoding the video itself. In other words, in this application the generality and cost advantages of going semantics-free outweigh the disadvantage of its semantic blindness. This fit argument is the fundamental reason this application was chosen.

6.4 Visualization of Temporal Behavior

To examine how the extraction results distribute over time, three diagnostic plots were generated for one complete extraction run: a timeline plot (Figure 6) marking candidate and finally selected frames along the video timeline; a coverage plot (Figure 7) checking whether the selected frames cover the whole video uniformly rather than piling up in one segment; and a gap plot (Figure 8) showing the distribution of intervals between adjacent selected frames, used to detect long empty stretches. Figures 9 and 10 show side-by-side comparisons of adjacent selected frames for manual verification of the deduplication stage — if compared frames remain highly similar, the dedup threshold is too loose. Together these diagnostics constitute inspectable evidence for the behavior of the scheme of Sections 6.1 and 6.2.

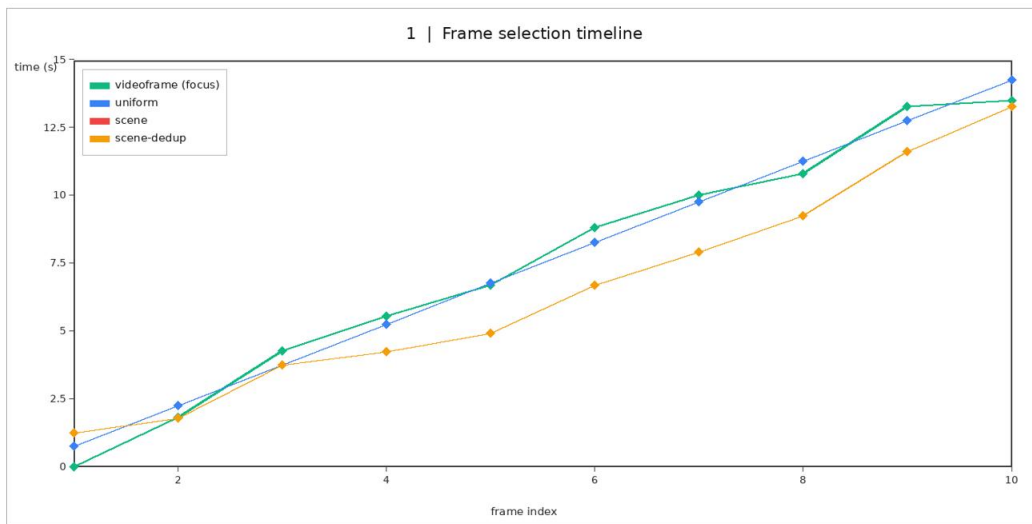


Figure 6 Output comparison of four frame-sampling algorithms along the video timeline. The horizontal axis is frame index; the vertical axis is time (s). The four curves correspond to videoframe (focus), uniform sampling, scene-change detection, and scene-change detection with perceptual deduplication.

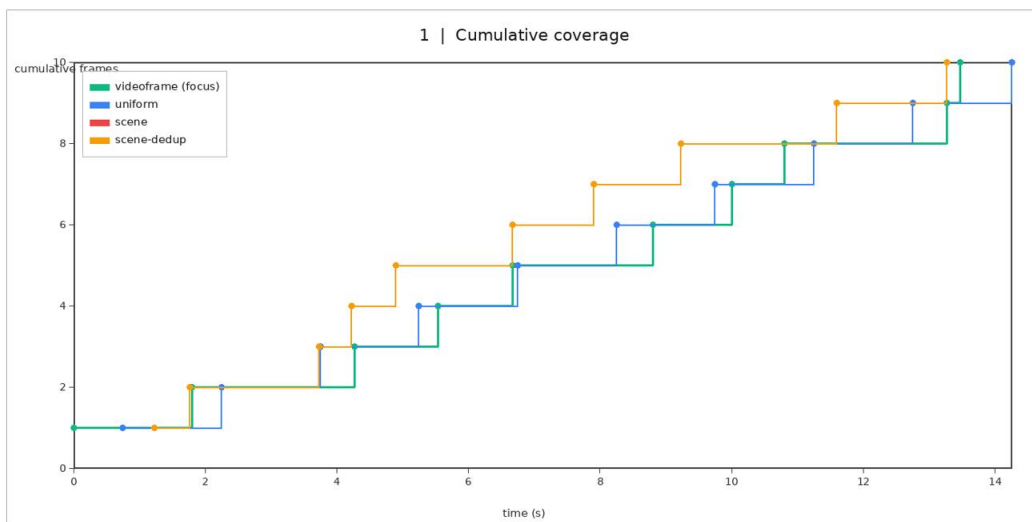


Figure 7 Temporal coverage of the video by the selected keyframes. (caption to be verified)

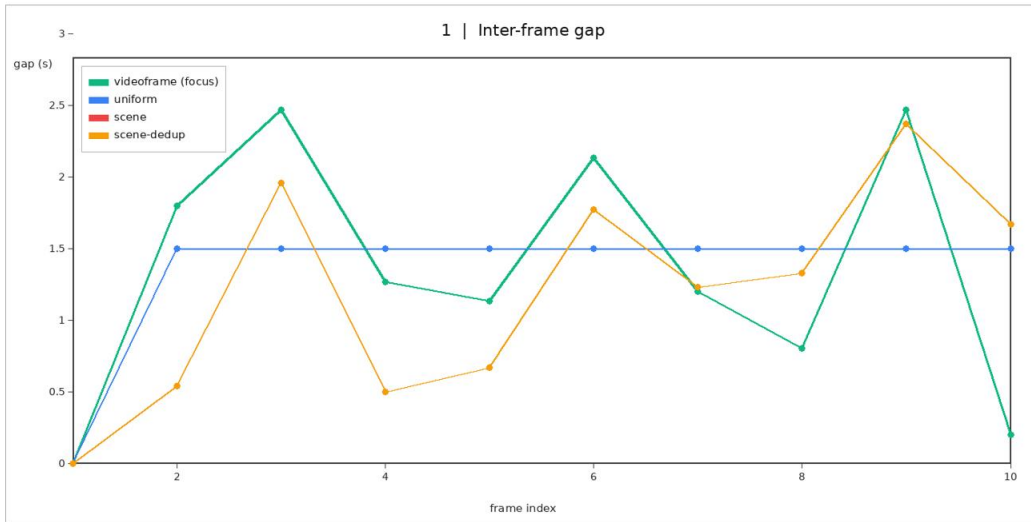


Figure 8 Distribution of time gaps between adjacent selected frames. (caption to be verified)

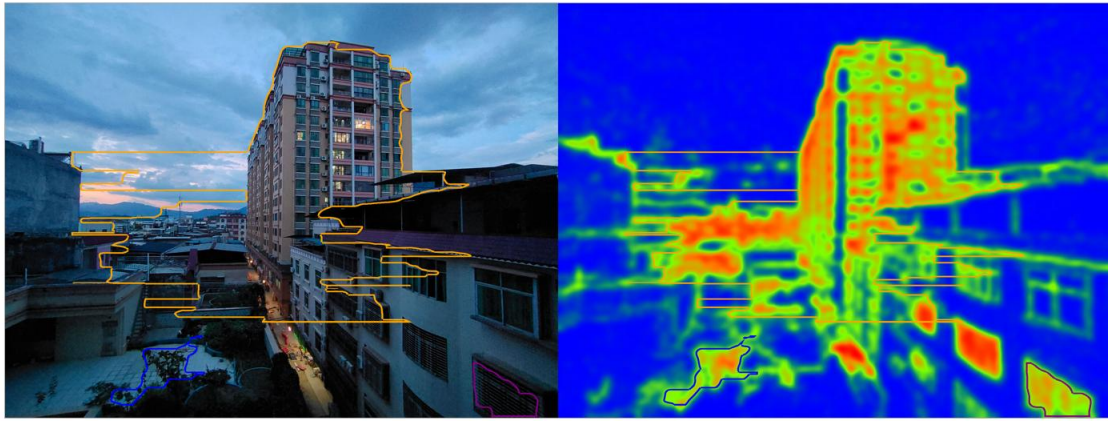


Figure 9 Saliency analysis of a single frame. Left: the original image with the outlines of the Top-3 focus regions overlaid (orange, blue, purple). Right: the corresponding pseudo-color heatmap, with the colormap running from blue (low saliency) to red (high saliency). Together the two panels constitute the complete visualization of the algorithm's output.

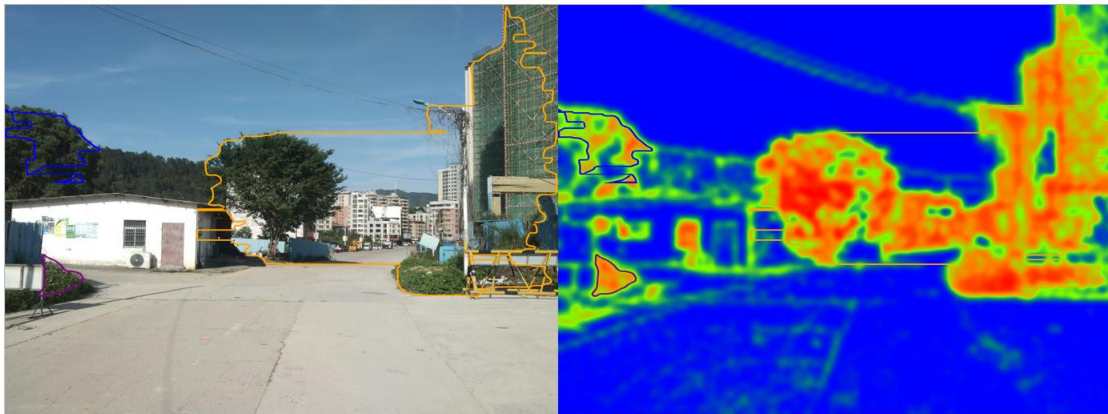


Figure 10 A second example of single-frame saliency analysis. Left: original image with focus regions overlaid; right: pseudo-color heatmap.

7 Conclusions and Future Work

This report presented a thoroughly semantics-free, bottom-up saliency algorithm whose core is the AND-style combination of edge density as the base channel with color rarity as a multiplicative gain, and applied it to video keyframe extraction. Qualitative observation

supports the following preliminary conclusion: the algorithm's output reliably falls within the human priority fixation set and can serve as a lower-bound estimate of the head of the fixation distribution; its training-free, cross-domain, explainable, linear-time character makes it particularly suitable for scenarios such as keyframe extraction, where generality matters more than semantic precision.

The report also enumerated eight known deficiencies in strict terms (Section 4). Future work, ordered by priority: first, complete quantitative validation on eye-tracking data under a proper protocol (Section 5.2), which is the precondition for every conclusion here to move from hypothesis to evidence; second, implement the three low-cost, high-benefit fixes — LoG preconditioning, quantile normalization, and the CIELAB conversion — with ablations; third, extend to multi-scale density; fourth, develop an absolute-threshold scheme for cross-image comparability. None of these changes the overall framework of the algorithm; all are incremental.

Appendix A Reproducibility Notes

The core implementation depends only on numpy and Pillow, with no GPU requirement. The exposed parameters number six: the color gain $\alpha = 4.0$; the strong-contour percentile of 99%; the block size of 16; the color-blobbing radius of 15; the macro smoothing radius of 10 together with UnsharpMask (2, 150, 3); and the focus threshold percentile of 80 with Top-3 selection. All parameters were held constant across all test domains, with no per-domain tuning — this is the premise under which the "semantics-free, cross-domain" claim holds, and also its falsifiable form: any refutation of the claim need only exhibit a single domain that works only with dedicated tuning.

The test images cited in this report and the runnable source code are released with the code repository; the diagnostic plots (Figures 6 through 10) are genuine outputs of the extraction pipeline.

Appendix B On Cost and Emergence

The extremely high complexity of this algorithm makes it inefficient and slow, yet it can describe the story of an entire video using as few images as possible. This suggests a lesson about emergence: what we call emergence, at the level of code, may amount to nothing more than an algorithm that merely imitates where a human would most likely focus in the absence of any context — and yet the order in which a story develops still falls out of it. In other words, the temporal order of a narrative need not be modeled explicitly; it may simply be the natural accumulation of per-frame fixation priority. A context-free, semantics-free local heuristic, extended along the time axis, spontaneously yields a coarse estimate of narrative order. This suggests that the conditions under which certain so-called higher-order capabilities emerge in code may be far more modest than we tend to assume.

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